

Spanning Tree–Induced Inequalities as Chvátal–Gomory Rank-1 Cuts

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1 Motivation and Setup

Optimization problems involving polynomials with binary variables arise in many areas of discrete optimization. A general binary polynomial optimization problem can be written as

$$\min_{z \in \{0,1\}^n} f(z) = \sum_{S \subseteq [n]} c_S \prod_{i \in S} z_i, \quad (1)$$

where the coefficients $c_S \in \mathbb{R}$ define the polynomial for every subset S of the n variables of the polynomial. Already in the quadratic case, where the objective contains only pairwise interactions, optimizing over binary variables is NP-hard [2]; allowing terms of higher degree only increases the combinatorial complexity of the problem.

These problems are usually tackled via a linearization by introducing an auxiliary variable for each multilinear term and enforcing consistency through a suitable system of linear constraints [1]. More precisely, using a hypergraph representation of monomials, let $G = (V, E)$ be a hypergraph. Consider the multilinear polytope, defined as the convex hull of all binary points $z \in \mathbb{Z}^{V \cup E}$ such that for every $e \in E$,

$$z_e = \prod_{v \in e} z_v.$$

The standard linearization of the multilinear polytope is given by the following linear constraints:

$$\begin{aligned} \alpha_v : \quad z_v &\leq 1 & \forall v \in V, \\ \nu_e : \quad -z_e &\leq 0 & \forall e \in E, \\ \varepsilon_e : \quad \sum_{u \in e} z_u - z_e &\leq |e| - 1 & \forall e \in E, \\ \delta_{v,e} : \quad -z_v + z_e &\leq 0 & \forall v \in e, e \in E. \end{aligned}$$

Relaxing the integrality condition $z \in \mathbb{Z}^{V \cup E}$ so that z is allowed to take fractional values in the above formulation yields the LP relaxation.

The polyhedral structure of the linearized formulation has been extensively studied: in the quadratic case, this yields the Boolean quadric polytope introduced and analyzed in [4].

Nonetheless, the LP relaxation highlighted above can be weak, exhibiting a large integrality gap between its optimal value and the optimal value of (1). Standard algorithms for these problems operate by solving successive LP relaxations of the integer formulation and their effectiveness depends critically on how closely the initial LP relaxation approximates the convex hull of integer-feasible solutions. Providing algorithms' formulations with a tight relaxation to start with is therefore fundamental.

Building on the work of [5], this dissertation aims to expand the family of valid inequalities used to tighten the LP relaxation. The central idea is to exploit hypergraph substructures to identify patterns that induce hyperplanes cutting off fractional solutions and thereby approximating the multilinear polytope more closely.

2 Main Results

2.1 Notations and Definitions

Definition 1 (Intersection graph). *Let $G = (V, E)$ be a hypergraph, and let $F \subseteq E$ be a nonempty subset of its hyperedges. The intersection graph induced by F in G , denoted $I_G(F)$, is the undirected graph*

$$I_G(F) = (V_I, E_I),$$

where

$$V_I := F,$$

$$E_I := \{ \{f', e'\} \subseteq F \mid f' \neq e', f' \cap e' \neq \emptyset \}.$$

Thus, each node of $I_G(F)$ corresponds to a hyperedge of G , and two nodes $f', e' \in F$ are adjacent in $I_G(F)$ if and only if the corresponding hyperedges intersect in G . For every $\{f', e'\} \in E_I$, we fix a representative intersection vertex $v_{f', e'} \in f' \cap e'$ to label the edge (f', e') of $I_G(F)$.

Definition 2 (Partial hypergraph). *Let $G = (V, E)$ be a hypergraph and let $E' \subseteq E$ be a subset of its hyperedges. The partial hypergraph induced by E' is*

$$H := (V', E'), \quad V' := \bigcup_{e \in E'} e.$$

In other words, H contains exactly the hyperedges in E' together with all vertices incident to them.

Definition 3 (Berge-cycle, γ -cycle, β -cycle). *Let $G = (V, E)$ be a hypergraph. A Berge-cycle of length $m \geq 2$ is a sequence*

$$v_1, e_1, v_2, e_2, \dots, v_m, e_m, v_1$$

such that

- $v_1, v_2, \dots, v_m$ are distinct vertices,
- $e_1, e_2, \dots, e_m$ are distinct hyperedges, and
- $v_i, v_{i+1} \in e_i$ for all $i = 1, \dots, m-1$, and $v_m, v_1 \in e_m$.

A hypergraph is Berge-acyclic if it contains no Berge-cycles.

A Berge-cycle is called a γ -cycle if $m \geq 3$ and each vertex v_i with $i = 2, \dots, m$ belongs exactly to the two consecutive cycle edges e_{i-1} and e_i , and to no other edge of the hypergraph.

A β -cycle is a γ -cycle with the additional requirement that also the vertex v_1 belongs only to the two edges e_1 and e_m , and to no other edge.

2.2 Spanning-Tree Inequality

Remark 1 (Assumptions). *Let $G = (V, E)$ be a hypergraph. Take $f \in E$ with $|f| \geq 2$ and let $F \subseteq E \setminus \{f\}$ be a nonempty subset of hyperedges. Consider the partial hypergraph*

$$H = (V(H), F \cup \{f\}), \quad \text{where } V(H) := \bigcup_{e \in F \cup \{f\}} e.$$

We assume that H contains a γ -cycle

$$C = v_1, e_1, v_2, e_2, \dots, v_m, e_m, v_1, \quad \text{with } e_1 = f.$$

Let T be any spanning tree of the intersection graph $I_H(F)$ induced by F .

Finally, we define the uncovered and covered vertices of f with respect to F as

$$P_f := f \setminus \bigcup_{e \in F} e, \quad C_f := f \setminus P_f.$$

For each $v \in C_f$, choose and fix one arbitrary hyperedge $e_v \in F$ such that $v \in e_v$. Such a choice is always possible by the definition of C_f . We denote by

$$E_{C_f} := \{e_v : v \in C_f\}$$

the collection of chosen hyperedges (note that the same hyperedge may appear multiple times).

Theorem 1 (Tree-spanning inequality). *With the notation and assumptions above, the inequality*

$$\boxed{\sum_{e \in F} z_e - \sum_{(f', e') \in T} z_{v_{f' e'}} - z_f + \sum_{v \in P_f} z_v \leq |P_f|} \quad (2)$$

is valid for the multilinear polytope and it is a Chvátal–Gomory cut of rank 1.

Proof. Set $\lambda := 1/|C_f|$. Take $(1 - \lambda) \cdot \nu_f$ and $\lambda \cdot \varepsilon_f$:

$$(1 - \lambda)(-z_f) + \lambda \left(\sum_{u \in f} z_u - z_f \right) \leq \lambda(|f| - 1).$$

resulting in

$$-z_f + \lambda \sum_{u \in f} z_u \leq \lambda(|f| - 1).$$

For each $v \in C_f$, add $\lambda \cdot \delta_{v, e(v)} : -\lambda z_v + \lambda z_{e(v)} \leq 0$ where $e_v \in E_{C_f}$ denotes the (fixed) hyperedge of F containing v (as defined in Remark 1, E_{C_f} is the set of chosen hyperedges associated with the vertices in C_f).

Summing over C_f yields

$$-z_f + \lambda \sum_{w \in P_f} z_w + \lambda \sum_{v \in C_f} z_{e(v)} \leq \lambda(|f| - 1).$$

For each $w \in P_f$ add $(1 - \lambda) \cdot \alpha_w : (1 - \lambda)z_w \leq (1 - \lambda)$. Summing over P_f raises each uncovered vertex from coefficient λ to 1 and contributes $(1 - \lambda)|P_f|$ to the right-hand side:

$$-z_f + \sum_{w \in P_f} z_w + \lambda \sum_{v \in C_f} z_{e(v)} \leq (1 - \lambda)|P_f| + \lambda(|f| - 1). \quad (3)$$

On a separate note, for each edge $(f', e') \in T$ we add the inequality

$$\delta_{v_{f' e'}, e'} : -z_{v_{f' e'}} + z_{e'} \leq 0,$$

and sum over all edges of T . Since each hyperedge $e \in F$ appears in the tree with degree $\deg_T(e)$ (i.e., the number of edges of T incident to e), this yields

$$\sum_{(f', e') \in T} (-z_{v_{f' e'}} + z_{e'}) = \sum_{e \in F} \deg_T(e) z_e - \sum_{(f', e') \in T} z_{v_{f' e'}} \leq 0.$$

For each $e \in F$ we now add $(\deg_T(e) - 1) \cdot \nu_e : (\deg_T(e) - 1)(-z_e) \leq 0$. Summing over all $e \in F$ normalizes the coefficients of the z_e -variables and gives

$$\sum_{e \in F} z_e - \sum_{(f', e') \in T} z_{v_{f' e'}} \leq 0.$$

Finally, for each $v \in C_f$ we add $\lambda \cdot \nu_{e(v)} : -\lambda z_{e(v)} \leq 0$, and summing over all $v \in C_f$ yields

$$\sum_{e \in F} z_e - \sum_{(f', e') \in T} z_{v_{f' e'}} - \lambda \sum_{v \in C_f} z_{e(v)} \leq 0. \quad (4)$$

Adding (3) to (4) yields

$$\sum_{e \in F} z_e - \sum_{(f', e') \in T} z_{v_{f' e'}} - z_f + \sum_{w \in P_f} z_w \leq (1 - \lambda)|P_f| + \lambda(|f| - 1).$$

Since $f = P_f \cup C$ where the union is disjoint, we have

$$|f| = |P_f| + |C_f|.$$

As $\lambda = 1/|C_f|$, the right-hand side equals

$$(1 - \lambda)|P_f| + \lambda(|P_f| + |C_f| - 1) = |P_f| + \lambda(|C_f| - 1) = |P_f| + \frac{|C_f| - 1}{|C_f|}.$$

Given the assumptions, $|C_f| \geq 2$ and therefore:

$$0 < \frac{|C_f| - 1}{|C_f|} < 1.$$

Therefore, taking the floor of the RHS gives exactly (2). All multipliers used before rounding are non-negative and since we only use inequalities of the standard linearization, (2) is a rank-1 CG cut. $\square$

2.3 Connections to Odd- β Cycle Inequalities

A subset of Odd- β cycle inequalities as part of the Spanning Tree inequality Odd β -cycle inequalities are defined in [5]. Some of them are contained within the family of spanning tree inequalities.

Theorem 2. *If G' contains odd β -cycles and the spanning-tree generating (2) is a path, then the inequality is an odd β -cycle inequality.*

Proof. The beta-cycle inequalities are defined in [5] as follows:

Consider a hypergraph $G = (V, E)$, let $C = v_1, e_1, v_2, \dots, v_m, e_m, v_1$ be a β -cycle in G , and let E^-, E^+ be a partition of $E(C)$ such that $k := |E^-|$ is odd and $e_1 \in E^-$. Let $D := \{e_{p+1}, e_{p+2}, \dots, e_m\}$, where e_p is the last edge in C that belongs to E^- . We denote by $f_1, \dots, f_k$ the subsequence of $e_1, \dots, e_m$ consisting of

the edges in E^- . Let $S_1 := (\cup_{e \in E^-} e) \setminus (\cup_{e \in E^+} e)$ and $S_2 := V(C) \setminus (\cup_{e \in E^-} e)$. With this notation in place, we make the following assumptions:

1. Every node $v \in \cup_{i=1}^m e_i$ is contained in at most two edges among $e_1, \dots, e_m$.
2. For every edge $e_i \in E^+ \setminus D$, every edge in E^- adjacent to e_i (if any) is either e_{i-1} or e_{i+1} .
3. No edge in D is adjacent to an edge f_i with i even.
4. At least one of the following two conditions holds:
 - (a) For every $v \in S_1$, either v is contained in just one edge $e \in E^-$, or it is contained in two edges f_i, f_j with i odd and j even.
 - (b) For every $e' \in E^-$ and $e'' \in D$ such that $e' \cap e'' \neq \emptyset$, then either $e' = e_1$, $e'' = e_m$ or $e' = e_p$, $e'' = e_{p+1}$.

We then define the odd β -cycle inequality corresponding to C and E^- as

$$\sum_{v \in S_1} z_v - \sum_{e \in E^-} z_e - \sum_{v \in S_2} z_v + \sum_{e \in E^+} z_e \leq |S_1| - |\{i \in \{1, \dots, m\} : e_i, e_{i+1} \in E^-\}| + \left\lfloor \frac{k}{2} \right\rfloor. \quad (5)$$

Before proceeding in the proof, it is necessary to check some notation: in the spanning tree construction, E^- coincides with f hence its cardinality k is always odd and equal to 1. $E(C) \setminus f$ is necessarily E^+ and D collapses to E^+ ; also, S_1 equals P_f . Finally, every hyperedge $e \in E^+$ appears as a node of the intersection graph, and two hyperedges p, e are connected if they share some vertex v_{pe} . If T is a path, the hyperedges $(f', e') \in E^+$ (that are now vertices in T) are adjacent in the β -cycle, and the intersection vertex $v_{f'e'} \in f' \cap e'$ is a vertex connecting them in the original hypergraph. Hence, the set

$$\{v_{f'e'} : (f', e') \in T\}$$

is precisely the set of vertices that: belong to the β -cycle (since they lie in two consecutive hyperedges of it) and do not belong to f . But those two conditions define S_2 exactly.

We now verify that assumptions (1)–(4a) from the definition of Odd- β -cycle inequalities are satisfied by the spanning tree construction introduced in Theorem 1.

Assumption (1). The β -cycle corresponding to the spanning tree inequality is generated by taking all hyperedges in $E(C) = E^- \cup E^+$, with $E^- = \{f\}$ and $E^+ = F$. In this case, the spanning tree T is a path. Therefore, every vertex appearing in the β -cycle is incident to at most two edges in the path, so assumption (1) is satisfied.

Assumption (2). By construction, $D = \{e_{p+1}, \dots, e_m\}$ collects the hyperedges of the cycle that appear after the last hyperedge in E^- . Since in our setting $E^- = \{f\}$ and $f = e_1$, we have $p = 1$ and consequently

$D = E^+ = E \setminus \{f\}$ the condition in assumption (2) becomes irrelevant, as $E^+ \setminus D$ is empty, and the adjacency requirement is trivially satisfied.

Assumption (3). Assumption (3) states that no hyperedge in D is adjacent to an edge f_i with even index. In our construction, $E^- = \{f_1\}$ consists of a single hyperedge, so $i = 1$ is odd and no even-indexed negative hyperedge exists. Therefore, assumption (3) is automatically fulfilled.

Assumption (4a). For assumption (4a) we have $S_1 = P_f$ and vertex $v \in P_f$ is, by definition, a vertex of f not contained in any other hyperedge of the cycle. Hence, every $v \in S_1$ belongs to exactly one hyperedge $e \in E^-$. This satisfies the first clause of assumption (4a), and thus assumption (4a) holds.

Consequently, the spanning tree (path) construction satisfies assumptions (1)–(4a) of the odd- β -cycle inequality definition. In this special case, the LHS resulting from the odd- β -cycle coincides with the spanning tree inequality (2), as shown by the identification $S_1 = P_f$, $S_2 = \{v_{f'e'} : (f', e') \in T\}$, $E^+ = F$ and $E^- = f$. The RHS of (5) shows 3 terms where:

1. $|S_1| = |P_f|$.
2. $|\{i \in \{1, \dots, m\} : e_i, e_{i+1} \in E^-\}| = 0$.
3. $\lfloor \frac{k}{2} \rfloor = \lfloor \frac{1}{2} \rfloor = 0$.

Thus the odd β -cycle inequality (5) coincides exactly with the spanning tree inequality (2) in this special case.

□

2.4 Subcycle and relation to facetness

We show that facet-defining spanning-tree inequalities are associated with maximal cycle structures. In particular, any spanning-tree inequality induced by a strict subcycle of a facet-inducing cycle cannot itself define a facet of the multilinear polytope.

Theorem 3. *Let $G = (V, E)$ be a hypergraph and fix a negative edge $f \in E$ with $|f| \geq 2$. Let $F \subseteq E \setminus \{f\}$ be such that the partial hypergraph induced by $F \cup \{f\}$ contains a γ -cycle with $e_1 = f$. Let T be a spanning tree of the intersection graph $I_G(F)$, and let $I(F, T)$ be the associated spanning-tree inequality. Assume that $I(F, T)$ is facet-defining for the multilinear polytope P .*

Let $F' \subset F$ be a proper subset such that the partial hypergraph induced by $F' \cup \{f\}$ still contains a γ -cycle with $e_1 = f$ and $P_f(F') \supset P_f(F)$. Let T' be the tree obtained from T by restricting it to the node set F' and repeatedly deleting leaves.

Then the spanning-tree inequality $I(F', T')$ is not facet-defining for P .

Proof. Let

$$F_2 := \{z \in P : A(F', T')(z) = b(F')\}$$

be the face of the multilinear polytope P induced by the tightness of the spanning-tree inequality $I(F', T')$. Here $A(F', T')(z)$ denotes the left-hand side of the inequality, viewed as a function of z whose coefficients depend on the sets F' and T' , while $b(F')$ denotes the right-hand side depending only on F' .

We prove that $\dim(F_2) < \dim(P) - 1$, hence $I(F', T')$ cannot be facet-defining.

Let

$$U := P_f(F') \setminus P_f(F), \quad U \neq \emptyset,$$

so that

$$b(F') = |P_f(F')| = |P_f(F)| + |U|.$$

Take any $z \in F_2$, i.e.

$$A(F', T')(z) = b(F').$$

Decompose the left-hand side as

$$A(F', T')(z) = \left(\sum_{e \in F'} z_e - \sum_{(e, e') \in T'} z_{v_{ee'}} \right) - z_f + \sum_{u \in P_f(F)} z_u + \sum_{u \in U} z_u.$$

For each edge $(e, e') \in T'$ we add the inequality

$$\delta_{v_{ee'}, e} : \quad -z_{v_{ee'}} + z_e \leq 0,$$

and sum over all edges of T' . Since each hyperedge $e \in F'$ appears in the tree with degree $\deg'_{T'}(e)$ (i.e., the number of edges of T' incident to e), this yields

$$\sum_{(e, e') \in T'} (-z_{v_{ee'}} + z_e) = \sum_{e \in F'} \deg_{T'}(e) z_e - \sum_{(e, e') \in T'} z_{v_{ee'}} \leq 0.$$

For each $e \in F'$ we now add $(\deg_{T'}(e) - 1) \cdot \nu_e : (\deg_{T'}(e) - 1)(-z_e) \leq 0$. Summing over all $e \in F'$ normalizes the coefficients of the z_e -variables and gives

$$\sum_{e \in F'} z_e - \sum_{(e, e') \in T'} z_{v_{ee'}} \leq 0.$$

and by ν_f we have $z_f \geq 0$. Hence

$$\left(\sum_{e \in F'} z_e - \sum_{(e, e') \in T'} z_{v_{ee'}} \right) - z_f \leq 0.$$

Moreover, by α_u we have $z_u \leq 1$ for all $u \in V$, so

$$\sum_{u \in P_f(F)} z_u \leq |P_f(F)|.$$

Combining the bounds yields

$$A(F', T')(z) \leq |P_f(F)| + \sum_{u \in U} z_u.$$

Since $z \in F_2$, we obtain

$$|P_f(F)| + |U| = A(F', T')(z) \leq |P_f(F)| + \sum_{u \in U} z_u,$$

and therefore

$$\sum_{u \in U} z_u \geq |U|.$$

Together with $z_u \leq 1$ this implies

$$z_u = 1 \quad \forall u \in U.$$

Hence

$$F_2 \subseteq P \cap \bigcap_{u \in U} \{z : z_u = 1\}.$$

Since $|U| \geq 1$, the equalities $z_u = 1$ for $u \in U$ are $|U|$ affinely independent constraints on P . Indeed, for every $u \in U$, the equality $z_u = 1$ is not valid for P (the all-zero feasible point of P satisfies $z_u = 0$), so these coordinate fixings reduce the dimension by at least $|U|$. Hence

$$\dim \left(P \cap \bigcap_{u \in U} \{z : z_u = 1\} \right) \leq \dim(P) - |U|.$$

We now show that the spanning-tree hyperplane $A(F', T')(z) = b(F')$ is not implied by these box constraints fixed at equality. Consider the binary point $\bar{z} \in P$ defined by

$$\bar{z}_v = 1 \quad \text{for all } v \in f, \quad \bar{z}_v = 0 \quad \text{for all } v \notin f,$$

and define $\bar{z}_e := \prod_{v \in e} \bar{z}_v$ for all $e \in E$. Then $\bar{z} \in P$, and since $U \subseteq f$, we have $\bar{z}_u = 1$ for all $u \in U$. We now evaluate the left-hand side of $A(F', T')$ at the witness point $\bar{z}$ defined by

$$\bar{z}_v = \begin{cases} 1 & \text{if } v \in f, \\ 0 & \text{if } v \notin f, \end{cases} \quad \bar{z}_e := \prod_{v \in e} \bar{z}_v \quad \forall e \in E.$$

Recall that

$$A(F', T')(z) = \left(\sum_{e \in F'} z_e - \sum_{(e, e') \in T'} z_{v_{ee'}} \right) - z_f + \sum_{v \in P_f(F')} z_v, \quad b(F') = |P_f(F')|.$$

We examine each term separately. As before:

$$\sum_{e \in F'} \bar{z}_e - \sum_{(e, e') \in T'} \bar{z}_{v_{ee'}} \leq 0.$$

Since every vertex of f has value 1 under $\bar{z}$, we have

$$\bar{z}_f = \prod_{v \in f} \bar{z}_v = 1,$$

and therefore

$$-\bar{z}_f = -1.$$

Because $P_f(F') \subseteq f$, each vertex $v \in P_f(F')$ satisfies $\bar{z}_v = 1$. Hence

$$\sum_{v \in P_f(F')} \bar{z}_v = |P_f(F')| = b(F').$$

By substituting the three bounds above into the definition of $A(F', T')$, we obtain

$$\begin{aligned} A(F', T')(\bar{z}) &= \left(\sum_{e \in F'} \bar{z}_e - \sum_{(e, e') \in T'} \bar{z}_{v_{ee'}} \right) - \bar{z}_f + \sum_{v \in P_f(F')} \bar{z}_v \\ &\leq 0 - 1 + b(F') \\ &= b(F') - 1. \end{aligned}$$

Since $b(F') - 1 < b(F')$, it follows that

$$A(F', T')(\bar{z}) < b(F'),$$

and therefore $\bar{z}$ is not tight for $I(F', T')$. □

3 Separation problem for Spanning-Tree Inequalities

Given a fractional point $\bar{z} \in \mathbb{R}^{V \cup E}$ obtained from the LP relaxation described in the setup, the separation problem for the spanning-tree inequalities introduced in Theorem 1 is the following:

Determine whether there exist $f \in E$, a nonempty family $F \subseteq E \setminus \{f\}$ of hyperedges, and a spanning tree T of the intersection graph $I_H(F)$ such that the corresponding spanning-tree inequality is violated by $\bar{z}$.

Equivalently, we seek f, F, T for which the violation

$$\psi(f, F, T; \bar{z}) := \sum_{e \in F} \bar{z}_e - \sum_{(f', e') \in T} \bar{z}_{v_{f'e'}} - \bar{z}_f + \sum_{v \in P_f} \bar{z}_v - |P_f| \quad (6)$$

is strictly positive. Here $P_f := f \setminus \bigcup_{e \in F} e$ denotes the uncovered vertices of f . A positive value of $\psi(f, F, T; \bar{z})$ means that the spanning-tree inequality (2) cuts off the fractional point $\bar{z}$.

The separation problem can be decomposed into two layers:

- (i) an *outer* combinatorial search over the choices of f and F , and
- (ii) an *inner* optimization over all spanning trees T of the intersection graph $I_H(F)$.

Only the second layer depends on the choice of the tree T . In order to isolate this dependence, we rewrite the violation in a form separating the contributions of f and F from that of T .

The last two terms of (6) can be grouped as

$$\sum_{v \in P_f} \bar{z}_v - |P_f| = \sum_{v \in P_f} (\bar{z}_v - 1),$$

so that

$$\psi(f, F, T; \bar{z}) = \sum_{e \in F} \bar{z}_e - \sum_{(f', e') \in T} \bar{z}_{v_{f'e'}} - \bar{z}_f + \sum_{v \in P_f} (\bar{z}_v - 1). \quad (7)$$

This shows that each uncovered vertex $v \in P_f$ contributes the term $(\bar{z}_v - 1)$ to the violation. In particular, all contributions of f and of the uncovered vertices P_f are independent of the choice of the spanning tree T .

For fixed f and F , the only term in (7) that depends on T is

$$- \sum_{(f', e') \in T} \bar{z}_{v_{f'e'}}.$$

3.1 Reduction to a minimum spanning tree

Given the above, we argue that maximizing $\psi(f, F, T; \bar{z})$ over all spanning trees T is equivalent to minimizing

$$S(T) := \sum_{(f', e') \in T} \bar{z}_{v_{f'e'}}.$$

This is exactly the problem of computing a minimum-weight spanning tree (MST) of the intersection graph $I_H(F)$, where each edge (f', e') has weight $w(f', e') := \bar{z}_{v_{f'e'}}$.

By the classical analysis of Kruskal's and Prim's algorithms (see [3, Chapter 4.5]), an MST can be computed in $O(|E_I| \log |F|)$ time. Since any two hyperedges in F may intersect, the intersection graph $I_H(F)$ can contain up to $\binom{|F|}{2}$ edges, that is, a number of edges quadratic in $|F|$. Consequently, computing an MST requires $O(|F|^2 \log |F|)$ time in the worst case. Let

$$S(\text{MST}(I_H(F))) = S(T^*) := \min_{T \text{ spanning tree of } I_H(F)} \sum_{(f', e') \in T} \bar{z}_{v_{f'e'}}.$$

Since $\psi(f, F, T; \bar{z})$ is maximized by the tree T^* achieving $\text{MST}(I_H(F))$, we obtain

$$\psi_{\max T}(f, F; \bar{z}) := \max_T \psi(f, F, T; \bar{z}) = \psi(f, F, T^*; \bar{z}), \quad (8)$$

that is,

$$\psi(f, F, T^*; \bar{z}) = \sum_{e \in F} \bar{z}_e - S(T^*) - \bar{z}_f + \sum_{v \in P_f} (\bar{z}_v - 1). \quad (9)$$

Therefore, for any fixed pair (f, F) , determining whether the best spanning-tree inequality is violated reduces to:

Compute T^* and check whether $\psi(f, F, T^*; \bar{z}) > 0$.

This establishes that the inner optimization over T (hence the test for violation for a fixed pair (f, F)) can be performed in polynomial time.

However, the full separation problem requires searching over all subsets $F \subseteq E \setminus \{f\}$; the cardinality of the latter is naively upper bounded by $|E| * 2^{|E|-1}$. Actually, it is important to observe that not every subset $F \subseteq E \setminus \{f\}$ is admissible: in order for a spanning-tree inequality to be well-defined, the partial hypergraph induced by $F \cup \{f\}$ must contain a γ -cycle. Thus the outer search does not range over all $2^{|E|-1}$ subsets, but only over those that participate in some γ -cycle with f . Nevertheless, the number of such cycle-inducing subsets may still be exponential in $|E|$, since a hypergraph may contain exponentially many distinct cycles. Consequently, even though the inner optimization over T is polynomial, the overall separation problem remains combinatorially large.

References

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